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Final Report

The Art of Scientific Computing: Project Title: High Dimensional Kalman Filters under Covariance Rotation

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1 Introduction

A Kalman filter (KF) is a type of algorithm which aims to estimate a hidden state in a linear system, given a series of observations. In particular, it handles the case in which the state variable evolves over time, and does so in a mean-squared optimal manner. This project aims to implement such an algorithm in the case of high dimensional linear regression, via the use of principle components analysis (PCA). The combination of PCA and KF exists in the literature, however, the papers largely deal with a one-time PCA fits, rather than updating representations over time. To extend this, we experiment with a new type of rotating covarinace structure, and test a way of keeping the filter in the correct space via a method we term *projection chaining*. We compare projection chaining to the usual static approach and discuss some properties and limitations of the method.

2 Background

We begin by providing background on the standard KF; its optimality and time complexity. We will present the theory in the case of linear regression by taking a special case of the original construction of [2]. Equation 1 is the state process equation which governs the dynamics of the unobserved state $\boldsymbol{\beta}_t \in \mathbb{R}^p$. Equation 2 is the measurement process, which describes the relationship between measurement $y_t \in \mathbb{R}$ and the observations $\boldsymbol{x}_t \in \mathbb{R}^p$.

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\varepsilon}_t \tag{1}$$

$$y_t = \boldsymbol{\beta}_t^T \boldsymbol{x}_t + u_t \tag{2}$$

where $\boldsymbol{\varepsilon}_t \sim N(0, \boldsymbol{Q})$, and $u_t \sim N(0, R)$. Here, $\boldsymbol{Q} \in \mathbb{R}^{p \times p}$ is the state noise covariance matrix, and R is the measurement noise (co)variance, both of which are assumed to be fixed over time in this project.

We denote the current estimate of $\boldsymbol{\beta}_t$ as of time $s \leq t$ as $\hat{\boldsymbol{\beta}}_{t|s}$. Further, we assume we have an initial state estimate $\boldsymbol{\beta}_{1|1}$; we describe a method of estimating this in Section 4. Define $\boldsymbol{P}_{t|s} = \mathbb{E}[(\boldsymbol{\beta}_t - \boldsymbol{\beta}_{t|s})(\boldsymbol{\beta}_t - \boldsymbol{\beta}_{t|s})^T] \in \mathbb{R}^{p \times p}$ to be the estimate error covariance.

At time t, we predict $\boldsymbol{\beta}_{t|t-1} = \boldsymbol{\beta}_{t-1|t-1}$, and $\boldsymbol{P}_{t|t-1} = \boldsymbol{P}_{t-1|t-1} + \boldsymbol{Q}$, then estimate what $\boldsymbol{y}_t$ would be under this state, given $\boldsymbol{x}_t$;

$$\hat{y}_t = \boldsymbol{\beta}_{t|t-1} \boldsymbol{x}_t$$

then calculate the innovation: $\tilde{y}_t = y_t - \hat{y}_t$. The central idea of the KF is that we assume $\boldsymbol{\beta}_{t|t} = \boldsymbol{\beta}_{t|t-1} + \boldsymbol{K}_t \tilde{y}_t$ where $\boldsymbol{K}_t$ is the *Kalman gain* at time t. Specifically, $\boldsymbol{K}_t$ is chosen to minimise $\mathbb{E}(\|\boldsymbol{\beta}_t - \boldsymbol{\beta}_{t|t}\|^2)$, and has closed form. Furthermore, we also update $\boldsymbol{P}_{t|t}$.

Indeed, a single prediction-update step for the standard KF has cubic time complexity in the state dimension, p [4]. As such, if p is large, this motivates the idea of perfoming our filtering in a q-dimensional space, where $q \ll p$, and then try to recover our state estimates in the original p-dimensional space. Indeed, the idea of running the KF in reduced-dimension PCA space is a proven method, as described in a recent review by Li et al. [3].

3 Dynamic Projections

Suppose we wish to update our PCA loadings as we progress through the iterations. This is a reasonable feature since the loadings estimates may become stale if we keep them static after initialisation. However, one issue arises from this; if we change our projection, then the usual update equations do not adapt to this. For example, the update $P_{k+1} = P_k + Q_k$ assumes the representation given by the k-th projection, likewise for updating the state estimate.

The first problem arises if we have a state $\boldsymbol{\beta}$, and assume we have an estimate of it in projected space: $\hat{\boldsymbol{\beta}}^U \approx \boldsymbol{\beta}U$, for $U \in \mathbb{R}^{p\times q}$, $q < p$. Then, suppose we wish to find the estimate in a different projected space, say, $\hat{\boldsymbol{\beta}}^V \approx \boldsymbol{\beta}V$, for $V \in \mathbb{R}^{p\times q'}$, $q' < p$. Additionally, suppose U and V arise from PCA so that $U^TU = I_q$ and $V^TV = I_{q'}$. Since we never observe $\boldsymbol{\beta}$ (or indeed, any estimate in the unprojected space), then we need an 'optimal' mapping from $\hat{\boldsymbol{\beta}}^U$ to $\hat{\boldsymbol{\beta}}^V$. We formalise this by the following optimisation problem:

$$\min_{W\in\mathbb{R}^{q\times q'}} \|\boldsymbol{\beta}^U W - \boldsymbol{\beta}^V\|_2$$

This is precisely a least squares problem, and the solution is

$$\begin{aligned} W &= ((\boldsymbol{\beta}^U)^T\boldsymbol{\beta}^U)^{-1}(\boldsymbol{\beta}^U)^T\boldsymbol{\beta}^V \\ &= (U^T\boldsymbol{\beta}^T\boldsymbol{\beta}U)^{-1}U^T\boldsymbol{\beta}^T\boldsymbol{\beta}V \\ &= U^TV \end{aligned}$$

So the least-square optimal state representation in V space is $\hat{\boldsymbol{\beta}}^V = \hat{\boldsymbol{\beta}}^U U^TV$. One can intuitively think of this as a mapping chain from $\mathbb{R}^q \xrightarrow{(1)} \mathbb{R}^p \xrightarrow{(2)} \mathbb{R}^{q'}$, where $(1 = U^T)$ tries to optimally 'undo' PCA and reconstruct the native space state, then $(2 = V)$ maps to $\mathbb{R}^{q'}$ via PCA.

We note that every time we apply an approximate projection, we permanently lose information in the direction of the bases lost in the prior projection. As such, we anticipate that this approach should not be chained for an extended number of iterations if the gross shift in the PCA loadings is expected to be very large.

3.1 Rotating Covariance

We propose a modification to the usual structure by allowing the PCA-loadings of the $\boldsymbol{x}$ data to rotate over time, while keeping the component variances (eigenvalues) constant. We can do this by using a block-rotation matrix R with reasonably small angles, and then we create the new covariance matrix via,

$$\Sigma' = R\Sigma R^T$$

Observe that implicitly we are rotating the PCA loadings since $R\Sigma R^T = RQ\Lambda Q^TR^T = Q'\Lambda(Q')^T$, where $Q' = RQ$. To be more precise, R is constructed as $R = \text{diag}(R(\theta_1), \ldots, R(\theta_k))$ for $k = p/2$ if p is even, else $R = \text{diag}(1, R(\theta_1), \ldots, R(\theta_{\lfloor p/2\rfloor}))$; see [1]. Hence, to rotate the data incrementally, we can start with some covariance matrix Σ_0, then define:

$$\Sigma_n = R^n\Sigma_0(R^n)^T$$

An important property of this transformation is that because the eigenvalues stay intact, the number of components which explain a given proportion of variance stays constant. In the algorithm, we commute R to produce some $R' = ORO^T$ for a random $O \in \mathbb{R}^{p\times p}$ orthogonal, thus randomising the planes for which we rotate about.

There is a question of how to best optimise this for a large dataset. One can directly construct R^n for each n by replacing $R(\theta_j)$ with $R(n\theta_j)$ for each $j = 1, \ldots, k$, hence eliminating the need to compute the matrix power. From there, we can batch multiply to calculate the entire sequence $(\Sigma_1, \ldots, \Sigma_N)$ simulataneously, provided memory constraints are respected. The construction of the block-diagonal rotation matrix can be computationally expensive if done individually for every n (i.e., every observation), so in the code we implement a more efficient version with numpy.einsum.

In context of the projection chaining apporach, perhaps it is viable if the accumulated subspace rotation across the dataset is small.

4 Dynamic PCA-KF

To initialise, we first run an initial projection, U_0, with PCA on some burn data, then calculate the reduced observations: $\boldsymbol{z} = \boldsymbol{x}U_0$. On these reduced q-dimensional observations, we estimate $\boldsymbol{\beta}'$ (the reduced state) to be the OLS estimate via the burn data: $\boldsymbol{\beta}'_{0|0} = (\boldsymbol{z}^T\boldsymbol{z})^{-1}\boldsymbol{z}^T\boldsymbol{y}$, then the measurement noise: $R = \frac{1}{n_{burn}}\sum_{t=1}^{n_{burn}}(y_t - \boldsymbol{\beta}_{0|0}^T\boldsymbol{z}_t)^2$, and by default, $\boldsymbol{P}_{0|0} = R(\boldsymbol{z}^T\boldsymbol{z})^{-1}$ (OLS estimate error covariance). Finally, we estimate the state noise covariance $\boldsymbol{Q}$ to be αI_q for some α chosen on a desired grid. We run ours on $\alpha \in \{10^{-9}, 10^{-8}, \ldots, 10^{-1}\}$ to span a wide gamut of scales.

This applies the previous projection chaining apporach to the PCA-KF algorithm. The algorithm is given as follows, and is run on every initialisation from the α grid:

1. **Predict:**
   (a) $\boldsymbol{P}_{t|t-1} = \boldsymbol{P}_{t-1|t-1} + \boldsymbol{Q}_{t-1|t-1}$
   (b) $\boldsymbol{\beta}'_{t|t-1} = \boldsymbol{\beta}'_{t-1|t-1}$
2. **Project (if not first iteration):**
   (a) Acquire the new PCA loadings $\boldsymbol{U}_t$.
   (b) Acquire the pushforward map: $\boldsymbol{V}_t = \boldsymbol{U}_{t-1}^T\boldsymbol{U}_t$
   (c) $\boldsymbol{P}_{t|t-1} \leftarrow \boldsymbol{V}_t^T\boldsymbol{P}_{t|t-1}\boldsymbol{V}_t$
   (d) $\boldsymbol{Q}_{t|t-1} \leftarrow \boldsymbol{V}_t^T\boldsymbol{Q}_{t|t-1}\boldsymbol{V}_t$
   (e) $\boldsymbol{\beta}'_{t|t-1} \leftarrow \boldsymbol{\beta}'_{t|t-1}\boldsymbol{V}_t$
   (f) $\boldsymbol{z}_t \leftarrow \boldsymbol{z}_t\boldsymbol{V}_t$: transform reduced observation vector.
   (g) $\boldsymbol{C}_t \leftarrow \boldsymbol{C}_t\boldsymbol{V}_t$; keep track of the accumulated projection.
3. **Update:**
   (a) $\delta_t = y_t - \boldsymbol{\beta}'^T_{t|t-1}\boldsymbol{z}_t$: innovation
   (b) $S_t = \boldsymbol{z}_t^T\boldsymbol{P}_{t|t-1}\boldsymbol{z}_t + R$: innovation covariance
   (c) $\boldsymbol{K}_t = S_t^{-1}\boldsymbol{P}_{t|t-1}\boldsymbol{z}_t$: Kalman gain
   (d) $\boldsymbol{\beta}'_{t|t} \leftarrow \boldsymbol{\beta}'_{t|t-1} + \delta_t\boldsymbol{K}_t$: $\mathbb{R}^q$ state estimate, $q \leq p$.
   (e) $\boldsymbol{\beta}_{t|t} \leftarrow \boldsymbol{\beta}'_{t|t}\boldsymbol{C}_t^T$; $\mathbb{R}^p$ state estimate.
   (f) $\boldsymbol{P}_{t|t} \leftarrow (I - \boldsymbol{K}_t\boldsymbol{z}_t^T)\boldsymbol{P}_{t|t-1}(I - \boldsymbol{K}_t\boldsymbol{z}_t^T)^T + R\boldsymbol{K}_t\boldsymbol{K}_t^T$: Joseph Update

We note that the predict and project steps can be swapped to no effect. The predict and update steps are generic to the KF algorithm when using the Joseph form of the update. We use Joseph form due to enhanced numerical accuracy compared to simpler forms of the update step [5].

To pick the best value of α from the grid, we first note that we should expect $S_t^{-\frac{1}{2}}\delta_t \sim N(0,1)$ under correct recovery of the state. So we pick the α which resulted in the process $(S_t^{-\frac{1}{2}}\delta_t)_t$ which is closest to being normally distributed, measured by the Shapiro-Wilk statistic* being closest to 1.

*https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.shapiro.html

5 Results and Analysis

Here we present some simulation results to test the speed and accuracy of the PCA-KF model. In addition, we provide a result which illustrates how 'stale' a static PCA estimate can get when the data is subjected to the covariance rotation as described in Section 3.1. Note that estimation error is calculated by measuring each of:

- The root mean squareed innovation: $\sqrt{\frac{1}{n-n_{burn}} \sum_{t=1}^{n-n_{burn}} \delta_t^2}$, and
- The root mean squared state error: $\sqrt{\frac{1}{n-n_{burn}} \sum_{t=1}^{n-n_{burn}} \|\boldsymbol{\beta}_{t|t} - \boldsymbol{\beta}_t\|^2}$

5.1 Dynamic Projection Tests

First we present a simulation result with the following hyperparamters:

- $(n, n_{burn}, p) = (2500, 400, 200)$, where n is the number of observations, p is the state dimension, and n_{burn} is the number of observations used to initialise PCA.
- Initial PCA proportion of variance retained (target): 0.80.
- Maximum angle increment (in radians) $\theta = 0.005$, or, $\frac{2\pi}{\theta} \approx 1257$.
- $\text{Var}(u_t) = 10^{-2}$
- ε_t construction scale parameter is 10^{-2}; this scales the $\boldsymbol{\varepsilon}$ covariance.
- x_t construction scale parameter is 10^{-2}; this scales the $\boldsymbol{x}$ covariance.

We fix an initial PCA rotation L, then at time t (of the non-burn data), we find the proportion of variance explained by L of the true covarinace Σ_t as: $\frac{\text{tr}(L^T \Sigma_t L)}{\text{tr}(\Sigma_t)}$.

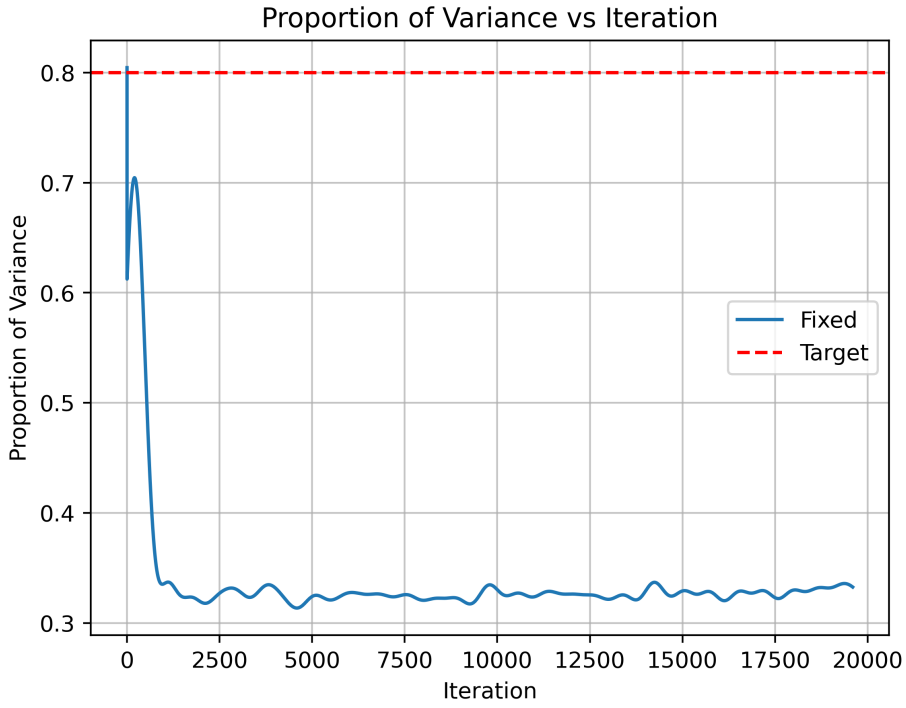


Figure 1: Proportion of variance retained by method.

We can observe from Figure 1 that the explained variance of the initial q eigenvectors of the true covariance matrix drops sharply. Indeed, past roughly 1000 iterations, the proportion of variance plateaus at ~ 0.33. This

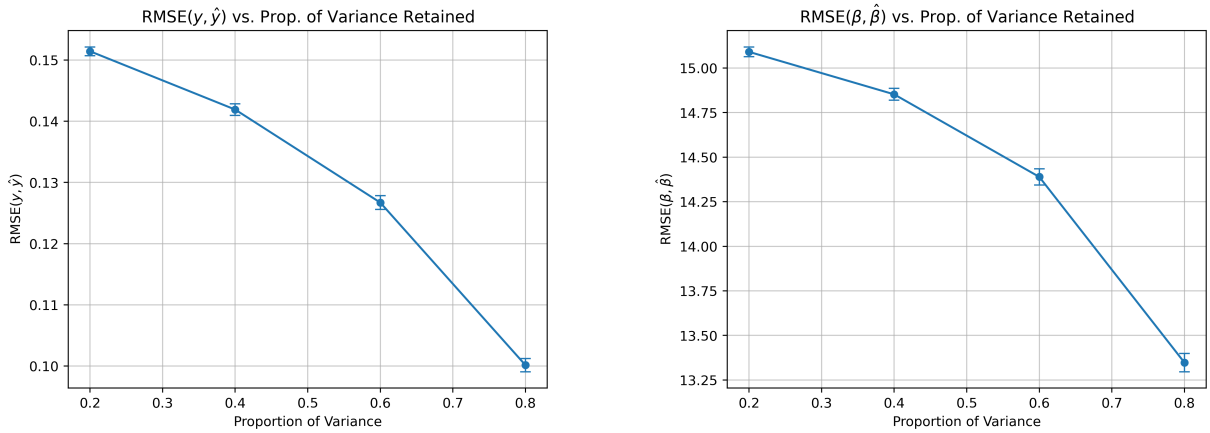
suggests the need to update our representations as we iterate through the algorithm. Interestingly, the plateau point is roughly where the maximum allowed period is, suggesting this is a useful heuristic cutoff point for where the projection gets stale.

5.2 Dynamic Filter

Now, we provide a simulation result for the full PCA-KF algorithm with the following hyperparamters:

- $(n, n_{burn}, p) = (2500, 500, 200)$
- Initial PCA proportion of variance retained (target): 0.90.
- Angle increment (in radians) $\theta = 0.0002$, to prevent excessive shrinkage of our estimate.
- $\text{Var}(z_t) = 10^{-2}$
- ε_t construction scale parameter is 10^{-2}
- x_t construction scale parameter is 10^{-2}

Figure 2: KF estimation error vs proportion of variance



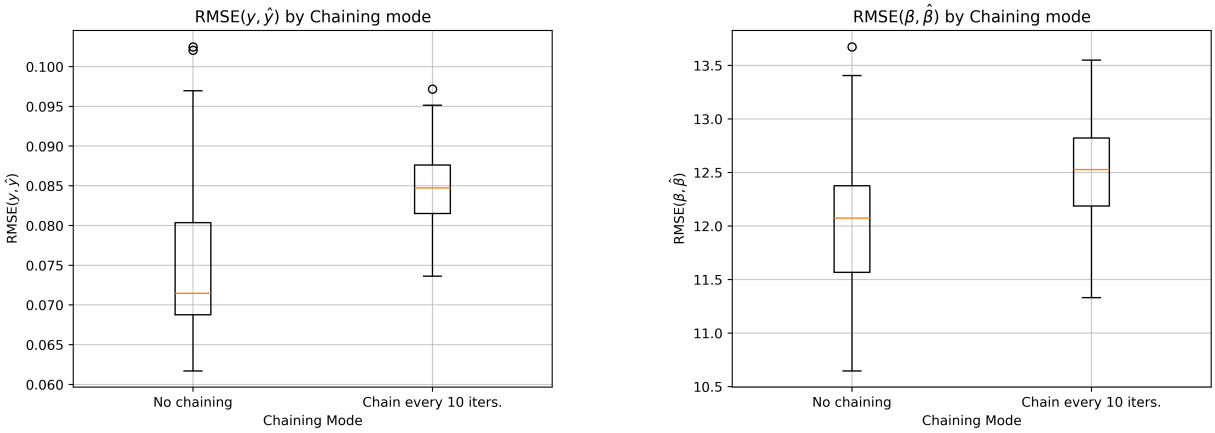
(a) RMSE($\hat{y}, y$)

(b) RMSE($\hat{\beta}, \beta$)

We see from Figure 2 that increasing the retained proportion of variance reduces the error in both metrics, well beyond the margin of error. However, the actual difference for the $\boldsymbol{\beta}$ RMSE is only increased by $\sim 13\%$ when going from 0.8 to 0.2 proportion of variance.

Finally, we calculate the fit times and number of components retained for $(n, n_{burn}) = (2000, 400)$, prop=0.90, same scale options as previous simulations, for $p \in \{50, 100, 200, 300\}$. The algorithm was run on an Intel i7 12700KF CPU.

Figure 3: KF estimation error vs chaining mode



(a) RMSE($\hat{y}, y$) **(b)** RMSE($\hat{\beta}, \beta$)

From Figure 3, we note that the chaining approach is convincingly less accurate than static regarding the innovation error, but less convincingly so for the state error. Ergo, we see no advantage to chaining projections over just keeping the static PCA. It appears that the loss of information across the iterations is sufficient that it worsens accuracy, despite it attempting to correct the drift in the projection of the static PCA. This may be inevitable if using any sort of projection chaining approach, since we are always right-multiplying within an existing space, meaning we can at best retain information over iterations. Indeed, our proposed projection chaining method is mean-squared optimal *within* the projection chaining framework. Perhaps worth noting is that chaining appeared to yield less dispersion in errors across different runs, despite having a higher mean error. This could be related to the 'shrinkage' of the estimate over time due to loss of subspace information.

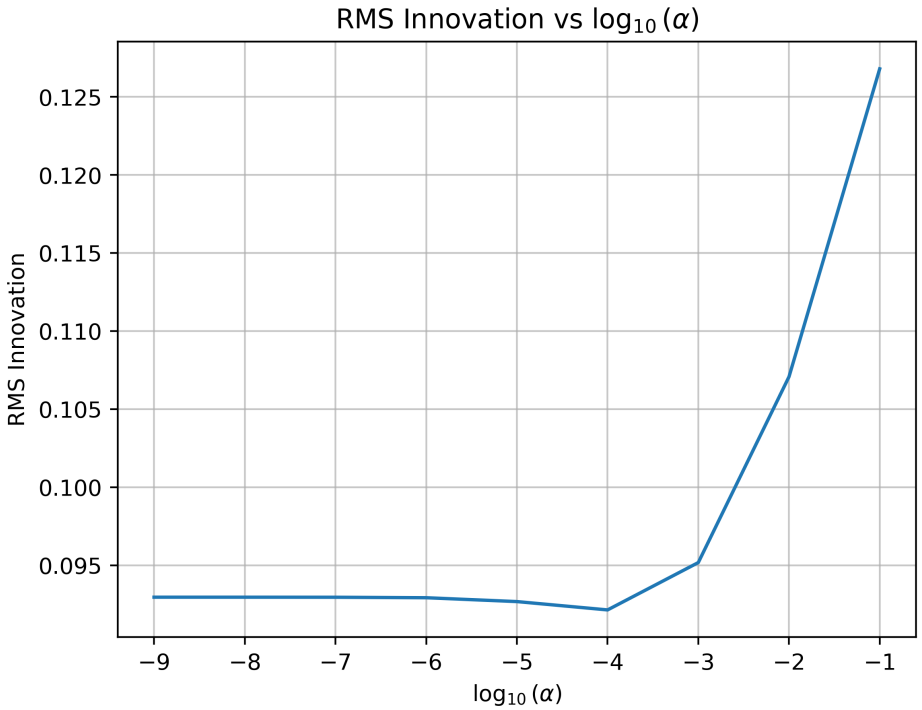


Figure 4: RMS Innovation vs. α

Figure 4 shows that the Q scaling factor affects the estimation accuracy greatly if it is too large. The idea behind this is that since $\boldsymbol{Q} = \text{Cov}(\boldsymbol{\beta}_t - \boldsymbol{\beta}_{t-1})$, then if we set this to be too large, our estimates become too sensitive to new updates, thus leading to excessive variance in our state estimates (hence large error). Indeed, this propogates to larger innovations $\delta_t = y_t - \hat{y}_t$. Interestingly, it appears difficult to make $\boldsymbol{Q}$ 'too small', at

least in the grid we chose. This is a useful robustness result, since we can only test so many values of $\boldsymbol{Q}$ before leading to computational intractability, thus defeating the purpose of the PCA-KF algorithm.

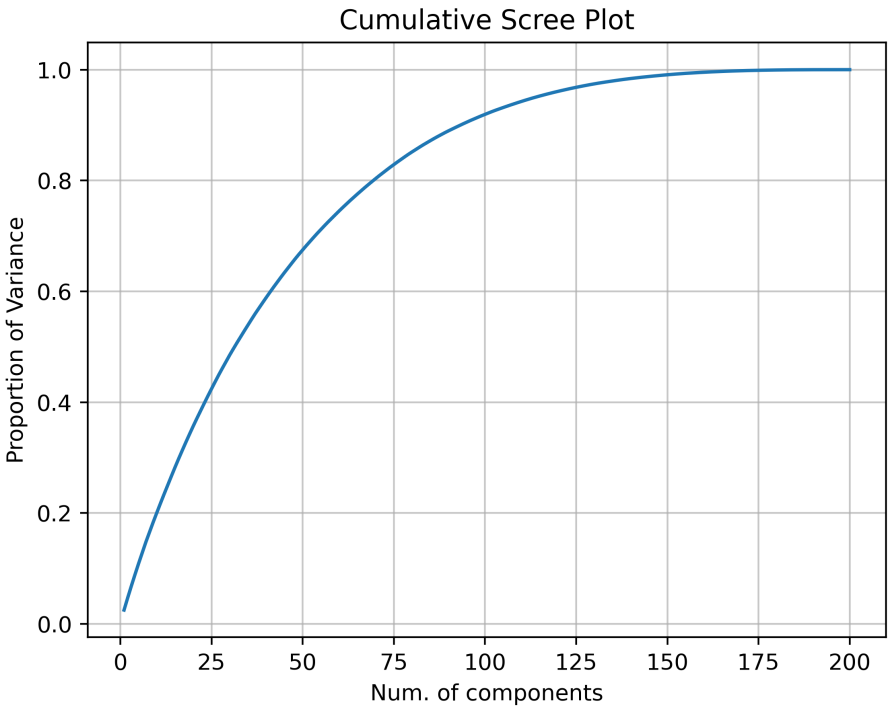
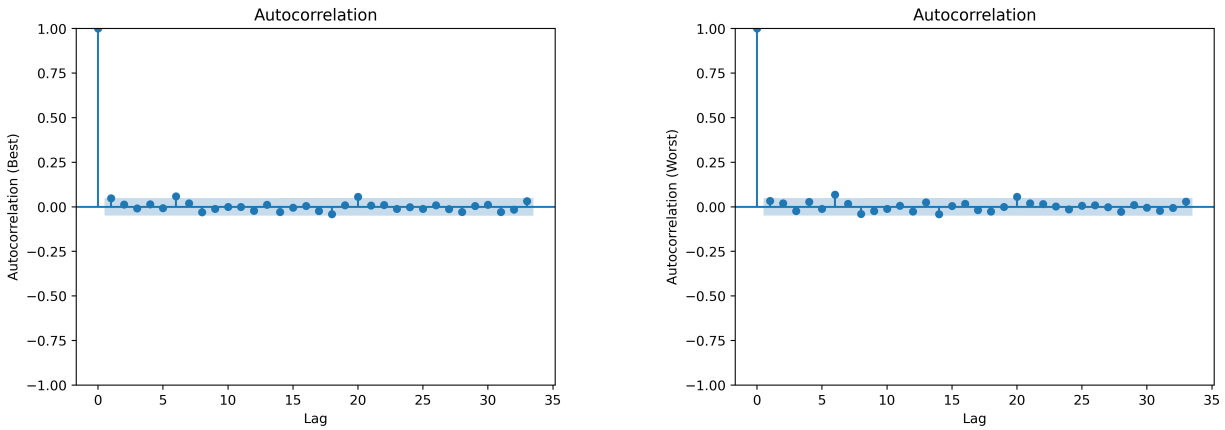


Figure 5: Cumulative Scree Plot for true initial covariance

Figure 5 illustrates the number of components required to retain a minimum level of variance retention. For example, at 50% of variance retained, we need only ~ 30 components out of the 200 total. We also observe that the 50 components with the least variance explained cumulatively explain very little variance; only $\sim 1-2\%$. So many components can of course greatly increase the runtime of the algorithm for almost negligble explanatory power, citing the earlier mentioned $\sim$cubic relationship between time complexity and the state dimension. A useful heuristic from this plot is that half of the components appear to explain $\sim 90\%$ of the variance.

Figure 6: Autocorrelation Function (ACF) Plots



(a) ACF for highest Shapiro-Wilk statistic

(b) ACF for lowest Shapiro-Wilk statistic

Figure 6 shows autocorrelation function (ACF) plots of the normalised innovations $(S_t^{\frac{-1}{2}}\delta_t)_t$ for both the 'worst' and 'best' values of α. The way to read it is that the value at Lag k shows the correlation between the process and itself, shifted k timesteps forward. We note that since we expect the process to be white noise, then one should also expect all the correlations for all lags (apart from 0) to be ~ 0. Curiously, this appears

to be the case for *both* the best and worse values of α, indicating robustness of the autocorrelation property to misspecification. Of couse, we also have the $N(0,1)$ property which needs to be satisfied, so the ACF does not fully capture the quality of the KF fit.

In Table 1, we run $n = 2000, n_{burn} = 400$, prop=0.80, and max rotation angle 0.001 for different values of p to measure the typical data generation + fit time.

Table 1: Runtime and Components retained: Average of 5 runs.

p	mean_time ± std. err. (s)	q
50	0.379 ± 0.001	17.2
100	0.945 ± 0.005	32.4
200	3.402 ± 0.010	59.2
300	6.378 ± 0.018	80.9

For a clean optimisation comparison, we can compare the fast block diag algorithm with the base one on $(n, p) = (3000, 200)$ at 0.80 prop. Averaging 10 calls to each function gives 0.839 ± 0.004s and 1.741 ± 0.007s for the fast and base algorithms respectively, where the error is the standard error. So we see the block diagonal matrix generation approximately halve in runtime under the numpy.einsum approach, indicating successful optimisation on that front.

6 Conclusion

1 AI Declaration

Claude Opus 5.5 was used to write an if statement to prepend a matrix with a value, and to generate an output filename in 'run_simulation.py'. Where this has been used, it has been labelled with a comment in the code.

2 Reproducing results

The simulations were run with the following hardware:

- **CPU**: Intel Core i7-12700KF (12 cores / 20 threads)
- **RAM**: 2 x 16 GB DDR4-3200
- **OS**: Windows 11 Pro (build 23600)

Note that in the following scripts, the seed is 123 by default (the same one used to generate the results). A seed can be set manually for 'rotation_test.py' and 'run_simulations.py' via the '–seed' argument.

Figure 1 can be reproduced via the following command:

```
python rotation_test.py --n 20000 --p 200 --burn 400 --prop 0.80 --max-angle 0.005 --ts
```

Figures 2 and 3 can be reproduced by first running the following simulation commands:

```
python run_simulation.py --n 2500 --p 200 --burn 500 --n-iters 200 --rotation-angle 0.0002 --
rotate-cov --ts --prop x
```

for $x \in \{0.2, 0.4, 0.6, 0.8\}$, and the following for the chain/no-chain test,

```
python run_simulation.py --n 2500 --p 200 --burn 500 --n-iters 50 --prop 0.90 --rotation-angle
0.0002 --rotate-cov --ts --prop 0.9
```

```
python run_simulation.py --n 2500 --p 200 --burn 500 --n-iters 50 --prop 0.90 --chain --chain-
every 10 --rotation-angle 0.0002 --rotate-cov --ts --prop 0.9
```

Then to generate the four plots, run:

```
python get_sim_plots.py
```

The data for Table 1 can be generated via running the following commands individually:

```
python run_simulation.py --n 2000 --p 50 --burn 400 --prop 0.80 --rotation-angle 0.001 --rotate-
cov --time --outfolder "data/time_benchmarks" --n-iters 5
python run_simulation.py --n 2000 --p 100 --burn 400 --prop 0.80 --rotation-angle 0.001 --rotate-
cov --time --outfolder "data/time_benchmarks" --n-iters 5
python run_simulation.py --n 2000 --p 200 --burn 400 --prop 0.80 --rotation-angle 0.001 --rotate-
cov --time --outfolder "data/time_benchmarks" --n-iters 5
python run_simulation.py --n 2000 --p 300 --burn 400 --prop 0.80 --rotation-angle 0.001 --rotate-
cov --time --outfolder "data/time_benchmarks" --n-iters 5
```

The data for the block diagonal comparison can be generated via the following command, and it prints to the terminal:

```
python run_simulation.py --n 3000 --p 200 --burn 400 --prop 0.80 --rotation-angle 0.001 --rotate-
cov --time --outfolder "data/time_benchmarks" --n-iters 1 --bench
```

Figures (...) are generated with the following command:

```
python run_simulation.py --n 2000 --p 200 --burn 400 --prop 0.80 --rotate-cov --rotation-angle
0.0001 --store-cov --get-additional-plots
```

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